

PETROLEUM PRODUCTION ENGINEERING COURSE EXERCISE



Title:	Electrical Submersible Pump Design
Major and Class:	Petroleum Engineering/ EN 2201
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《Petroleum engineering》 Course design assignment

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Name	Dewan Shiropa	Student No.	Lx2213010218	Major and Class	Petroleum Engineering/ EN 2201
Assignments	1. Data given: Well depth: 10,000-ft Oil gravity: 40 °API GOR: 50 scf/stb Water cut: 0 Tubing ID: 2.922-in Casing ID: 7 in Bo: 1.25 μ_o: 5cp γ_g: 0.7 T_{surface}: 70°F $T_{\text{bottomhole}}$: 170°F Pr: 4350 psia AOF: 15,000stb/day P_{wh}: 100psi $P_{\text{suction-min}}$: 200psi $P_{\text{discharge}}$: 4000psi 2. Design assignment q_{ld}=10000stb/day, Select an ESP, determine the number of stages and required power.				
Beginning and ending time		9th April, 2025---30th April, 2025			
Supervisor Signature					
Student Signature					

1. Oil Production and Artificial Lift Methods

Oil production is the process of extracting hydrocarbons from subsurface reservoirs and bringing them to the surface for processing and distribution. During the early life of a well, production typically occurs under natural reservoir pressure, allowing fluids to flow freely. However, as reservoir pressure declines over time, the natural energy becomes insufficient to sustain economical production rates. To overcome this, artificial lift methods are employed to enhance the flow of oil from the reservoir to the surface.

Artificial lift is a vital technique in the petroleum industry used to increase production from low-pressure or depleted wells. It involves using mechanical means to reduce bottom-hole pressure, thereby encouraging fluid flow. There are several types of artificial lift methods, each suitable for different well conditions.

The most common artificial lift methods include **rod pumping (beam pumping)**, **gas lift**, **electric submersible pumps (ESPs)**, **progressing cavity pumps (PCPs)**, and **hydraulic piston pumps**. Beam pumping systems use a surface motor and sucker rods to move a downhole pump. Gas lift involves injecting gas into the production tubing to reduce the fluid density and improve flow. ESPs use a downhole electric motor to drive a multistage centrifugal pump and are effective in high-volume wells. PCPs are suitable for viscous fluids and involve a rotor-stator assembly that moves fluids upward.

Choosing the right artificial lift method depends on various factors including production rate, fluid properties, well depth, and operational costs. Effective artificial lift ensures maximum reservoir recovery, improved production rates, and extended well life. As technology advances, these systems continue to evolve, offering more efficient and intelligent lifting solutions to meet the growing energy demand.

2. Electric Submersible Pump (ESP) System

Artificial lift is essential for boosting hydrocarbon production from wells when reservoir pressure is no longer sufficient to push fluids to the surface. Among the various artificial lift methods, the **Electric Submersible Pump (ESP)** is one of the most widely used, especially for high-volume wells. ESP systems are designed to efficiently lift large volumes of fluid from great depths and are ideal for both offshore and onshore oil fields.

Components of an ESP System

An ESP system consists of several key components installed in a series, forming a continuous unit that operates while submerged in the production fluid. The main components are:

Pump

The pump is a **multistage centrifugal pump** designed to handle high fluid volumes. It is composed of impellers and diffusers:

- **Impellers** impart kinetic energy to the fluid.
- **Diffusers** convert kinetic energy into pressure energy.

The number of stages depends on the required total dynamic head (TDH) to lift fluids.

Motor

The motor is a **three-phase induction electric motor** placed directly below the pump. It is filled with a dielectric oil to lubricate and cool internal parts. The motor is sealed to prevent fluid ingress and designed to operate under extreme downhole temperatures and pressures.

Seal (Protector)

The protector or seal chamber is critical to prevent well fluids from entering the motor. It equalizes the internal pressure of the motor with the wellbore pressure and compensates for volume changes due to temperature.

Intake and Gas Separator

- **Intake** allows fluid entry from the wellbore to the pump.
- A **gas separator** may be used to remove free gas before fluid enters the pump. Gas in the pump can cause inefficiency or failure due to gas locking.

Power Cable

A specially insulated cable transmits electrical power from the surface control unit to the downhole motor. It must withstand high temperatures, pressures, and corrosive environments.

Surface Equipment

This includes the **Variable Speed Drive (VSD)**, **Transformer**, and **Motor Controller**. The VSD adjusts the motor's speed to optimize pump performance and protect the system against voltage fluctuations and overloads.

Working Principle of ESPs

The Electric Submersible Pump operates on the principle of **centrifugal force**. Here's a step-by-step explanation of the operation:

1. **Fluid Entry:** Fluid from the reservoir enters the pump through the intake section.
2. **Impeller Rotation:** The motor rotates the impellers at high speed (typically 3,500 to 4,200 RPM). Each impeller adds kinetic energy to the fluid.

3. **Pressure Buildup:** As the fluid moves through the diffusers, the kinetic energy is converted into pressure. Multiple stages are used to build sufficient head to lift the fluid to the surface.
4. **Gas Separation (if applicable):** Free gas is separated before fluid enters the pump, improving efficiency.
5. **Fluid Lifting:** The pressurized fluid is transported up the tubing to the surface.

This entire system is controlled by a surface unit that can adjust the motor's operating parameters based on real-time production needs.

Advantages of ESP Systems

The Electric Submersible Pump system offers several advantages over other artificial lift methods, making it a popular choice for various well conditions:

High Production Rates

ESPs are capable of handling **very high volumes** (up to 30,000+ stb/day), making them ideal for high-yielding wells.

Deep Well Applications

Due to their design, ESPs can operate at **greater depths** (over 15,000 feet), where rod pumps or gas lifts may not be practical.

Suitability for Deviated or Horizontal Wells

Unlike rod pumps, ESPs have no reciprocating motion and are more suitable for **directional or horizontal wells**.

Low Surface Footprint

ESPs require **minimal surface equipment**, making them particularly suitable for offshore platforms or remote locations.

Automation and Control

Modern ESP systems can be integrated with **digital monitoring and control**, allowing real-time optimization of pump performance using VSDs and sensors.

Low Noise and Vibration

Compared to surface-mounted lift systems, ESPs produce **less surface noise and mechanical vibration**.

Disadvantages and Limitations

While ESPs have many advantages, they also come with certain challenges:

- **High Initial Cost:** ESP systems are relatively expensive to install and replace.
- **Power Requirements:** They need a **reliable power source** and may incur high energy costs.
- **Gas Handling Limitations:** ESPs may experience **gas locking** if too much free gas enters the pump.
- **Corrosion and Scaling:** The presence of corrosive fluids or scale-forming minerals can reduce equipment life.
- **Maintenance Difficulty:** Any failure typically requires a **workover**, which increases downtime and cost.

New Technologies in Electric Submersible Pumps

With advances in digital oilfield technologies and materials engineering, ESP systems have seen numerous innovations. Some key developments include:

Variable Speed Drives (VSDs)

Modern ESP systems are increasingly equipped with VSDs, which:

- Adjust motor speed to optimize performance
- Reduce energy consumption
- Provide soft-start functionality to extend pump life
- Respond to changing well conditions automatically

Permanent Magnet Motors (PMMs)

PMMs are more efficient and compact than traditional induction motors. Benefits include:

- Reduced heat generation
- Increased system efficiency (up to 15% better)
- Improved torque at low speeds

Advanced Gas Handling Devices

New designs in gas separators and intake systems significantly improve ESP performance in gassy environments. Technologies like:

- **Cyclonic separators**
- **Gas-handling impellers**
help reduce gas interference and prevent gas locking.

Real-Time Monitoring and IoT Integration

Advanced ESPs now come with downhole sensors and are integrated into digital platforms for:

- Monitoring intake and discharge pressure
- Measuring motor temperature and vibration
- Predicting failure using AI and machine learning
- Optimizing lift in real-time based on production analytics

High-Temperature and High-Solids ESPs

New materials and coatings are being used to develop **high-temperature ESPs** (HT-ESPs) for geothermal or HPHT environments. Others are being tailored for **sandy or high-solids wells**, with abrasion-resistant components.

Cableless ESPs

Innovative systems like **cableless ESPs** (e.g., Novomet's rigless ESP) use coiled tubing deployment and avoid traditional power cables. These systems:

- Reduce rig time and intervention cost
- Enable quicker installation and retrieval
- Are ideal for marginal or offshore wells

Applications of ESPs

ESPs are widely used in:

- **Offshore production:** Compact design suits platform constraints.
- **Heavy oil production:** With proper motor cooling and PCP combinations.
- **Enhanced oil recovery (EOR):** For fields with declining pressure.
- **Unconventional reservoirs:** Including shale and tight oil, when gas lift isn't viable.

Procedure of ESP Selection

The selection of an Electric Submersible Pump (ESP) is a critical step in ensuring efficient artificial lift in oil production. A properly selected ESP can maximize production, minimize downtime, and extend equipment life. The selection procedure involves evaluating both reservoir characteristics and surface constraints to determine the best pump configuration. Below is a step-by-step guide to the standard procedure of ESP selection:

1. Data Collection

Begin by gathering **basic well and reservoir data**, which are essential for sizing and selecting the ESP. This includes:

- **Well depth** (measured depth and true vertical depth)
- **Reservoir pressure (Pr)** and **temperature**
- **Flowing bottom-hole pressure**
- **Well production rate (target and current)**
- **Fluid properties** such as oil gravity, viscosity, water cut, GOR (Gas Oil Ratio), and formation volume factor (Bo)
- **Tubing and casing size**
- **Gas content and potential for gas interference**
- **Sand or solids content**

This data provides the foundation for accurately calculating the head requirements and selecting components suited to the operating environment.

2. Determine Required Pump Capacity

Based on the desired liquid production rate (qld), determine the **required pump capacity**. Consider not only the current production but also the anticipated future range due to reservoir decline or stimulation.

3. Total Dynamic Head (TDH) Calculation

Calculate the **total dynamic head** the pump must overcome to lift the fluid to the surface. This includes:

- Vertical lift (well depth adjusted for fluid gradient)
- Friction losses (in tubing)
- Wellhead back pressure
- Safety margins
- Intake pressure to avoid cavitation or gas lock

Convert this head into **pressure (psi)** and divide by the pressure per stage of the pump to determine the **number of stages** required.

4. Pump and Motor Selection

Choose a pump that matches the flow rate and head requirement. The manufacturer's pump performance curves help in selecting the appropriate **pump series and number of stages**.

Select a **motor** with enough power (HP) to drive the pump at the desired conditions. Consider:

- Motor load (brake horsepower)
- Operating voltage
- Motor cooling
- Surface power supply limitations

5. Gas Handling Consideration

If free gas at the pump intake exceeds 10-15%, include **gas separators or gas handlers**. This helps prevent **gas locking**, which reduces pump performance.

6. Cable and Seal Section Selection

Choose an **ESP power cable** that can carry the required electrical load and withstand wellbore conditions (temperature, fluid composition). The **seal section (protector)** should accommodate thermal expansion and protect the motor from well fluids.

7. Check Compatibility and Review

Ensure all components—pump, motor, intake, protector, gas handler, cable, and surface controller—are **compatible** in size and performance. Review the design against expected changes in reservoir and well performance.

8. Final System Design and Optimization

Finalize the ESP configuration, and use **software simulation tools** (like Prosper or AutographPC) to optimize the system's performance under various scenarios. Adjust parameters for efficiency, energy consumption, and longevity.

Procedure of ESP selection

Vogel IPR — Calculate Pump Intake Pressure

Use Vogel's equation:

$$\frac{q}{q_{max}} = 1 - 0.2 \left(\frac{P_{wf}}{P_r} \right) - 0.8 \left(\frac{P_{wf}}{P_r} \right)^2$$

Where:

- $q=10,000$ stb/day
- $q_{max}=12,000$ stb/day
- $P_r=4350$ psia

Solve for P_{wf} :

$$\frac{10000}{12000} = 1 - 0.2x - 0.8x^2 \rightarrow 0.8333 = 1 - 0.2x - 0.8x^2$$

$$= 0.8x^2 + 0.2x - 0.1667 = 0$$

$$= x = \frac{-0.2 \pm \sqrt{0.2^2 - 4(0.8)(-0.1667)}}{2(0.8)} = \frac{-0.2 \pm \sqrt{0.04 + 0.5334}}{1.6} = 0.348269$$

$$= 0.348269 \times 4350 = 1515 \text{ psia (ESP intake pressure)}$$

Total Dynamic Head (TDH)

We calculate TDH needed to raise fluid from 10,000 ft to surface and overcome pressure drop:

$$TDH = H_{\text{lift}} + H_{\text{surface backpressure}} + H_{\text{friction (est.)}}$$

$$H_{\text{lift}} = 10,000 \text{ ft}$$

b. Surface Pressure Head

Convert surface WHP (100 psia) to head:

We need fluid **density** first:

- Oil Gravity = 40 °API
- $SG \approx \frac{141.5}{40 + 131.5} = 0.825$

$$\rho = \frac{62.4 \times SG}{Bo} = \frac{62.4 \times 0.825}{1.25} \approx 41.14 \text{ lb/ft}^3$$

$$H_{\text{surface}} = \frac{144 \cdot 100}{41.18 \cdot 32.2} = 10.86 \text{ ft}$$

c. Friction Head Loss (Estimate)

Approximate for tubing = ~800 ft

TDH Total:

$$TDH = 10000 + 10.86 + 800 = 10,810.86 \text{ ft}$$

Power Requirement

$$\text{Pump HP} = \frac{0.00273 \cdot q \cdot TDH \cdot P}{\eta}$$

Where:

- $q = 10000/24 \approx 416.67 \text{ stb/hr}$
- $\rho = 41.14 \text{ lb/ft}^3$
- $TDH = 10,908 \text{ ft}$
- $\eta = 0.6$ (typical assumed efficiency)

$$\text{Pump HP} = \frac{0.00273 \cdot 416.67 \cdot 10810.86 \cdot 41.14}{0.6}$$

$$= 843,190.25 \text{ HP}$$

Estimate Number of Stages

Typical head per stage $\approx 60 \text{ ft}$ (depends on viscosity and flow rate)

$$\text{Stages} = 10,810.86/60 \approx 180 \text{ stages}$$

Final ESP Design Summary

Specification	Value
Flow Rate	10,000 stb/day
Pump Intake Pressure	1515 psia
Discharge Pressure	$\geq 4,000 \text{ psia}$
TDH	10,810.86 ft
Horsepower Required	843,190.25 HP
Pump Stages	$\sim 180 \text{ stages}$
Fluid Temp (Intake)	170 °F
Fluid Gravity (SG)	~ 0.825